

Python DSA - Useful Functions & Methods

1. LIST METHODS

`a.sort()` — Sorts the original list.

```
Example:  
a = [3,1,2]  
a.sort()  
print(a)  
Output: [1, 2, 3]
```

`sorted(a)` — Returns a new sorted list; original stays unchanged.

```
Example:  
a = [3,1,2]  
print(sorted(a))  
Output: [1, 2, 3]
```

`a.reverse()` — Reverses the original list.

```
Example:  
a = [1,2,3]  
a.reverse()  
print(a)  
Output: [3, 2, 1]
```

`a.append(x)` — Adds one element at the end.

```
Example:  
a = [1,2]  
a.append(3)  
print(a)  
Output: [1, 2, 3]
```

`a.extend([x,y])` — Adds multiple elements.

```
Example:  
a = [1,2]  
a.extend([3,4])  
print(a)  
Output: [1, 2, 3, 4]
```

`a.insert(index, value)` — Inserts a value at a specific index.

```
Example:  
a = [1,3]  
a.insert(1,2)  
print(a)  
Output: [1, 2, 3]
```

`a.remove(x)` — Removes the first matching value.

```
Example:  
a = [1,2,3]  
a.remove(2)  
print(a)  
Output: [1, 3]
```

`a.pop()` — Removes and returns the last element.

```
Example:  
a = [1,2,3]  
x = a.pop()  
print(x)  
Output: 3
```

`a.count(x)` — Counts how many times x occurs.

```
Example:  
a = [1,2,2,3]  
print(a.count(2))  
Output: 2
```

`a.index(x)` — Returns the first index of x.

```
Example:  
a = [10,20,30]  
print(a.index(20))  
Output: 1
```

`x in a` — Checks whether x exists in the list.

```
Example:  
a = [10,20,30]  
print(20 in a)  
Output: True
```

2. STRING METHODS

`s.isalnum()` — True only when all characters are letters or digits.

```
Example:  
print("Hello123".isalnum())  
Output: True
```

`s.isalpha()` — True only when all characters are letters.

```
Example:  
print("Hello".isalpha())  
Output: True
```

`s.isdigit()` — True only when all characters are digits.

```
Example:  
print("123".isdigit())  
Output: True
```

`s.islower()` — Checks whether cased letters are lowercase.

```
Example:  
print("hello".islower())
```

Output: True

`s.isupper()` — Checks whether cased letters are uppercase.

Example:

```
print("HELLO".isupper())
```

Output: True

`s.lower()` — Converts letters to lowercase.

Example:

```
print("HELLO".lower())
```

Output: hello

`s.upper()` — Converts letters to uppercase.

Example:

```
print("hello".upper())
```

Output: HELLO

`s.strip()` — Removes whitespace from both ends.

Example:

```
print(" hello ".strip())
```

Output: hello

`s.replace(old, new)` — Replaces occurrences of old with new.

Example:

```
print("a-b".replace("-", ""))
```

Output: ab

`s.split()` — Splits a string into a list.

Example:

```
print("hello world".split())
```

Output: ['hello', 'world']

`' '.join(a)` — Joins elements using the separator.

Example:

```
print(" ".join(["hello", "world"]))
```

Output: hello world

`s.find(x)` — Returns the first index of x; -1 if not found.

Example:

```
print("hello".find("l"))
```

Output: 2

3. BUILT-IN FUNCTIONS

`max(a)` — Returns the largest value.

Example:

```
a = [4,9,2]
```

```
print(max(a))
```

Output: 9

`min(a)` — Returns the smallest value.

Example:

```
a = [4,9,2]
```

```
print(min(a))
```

Output: 2

`sum(a)` — Returns the total of numeric elements.

Example:

```
a = [1,2,3]
```

```
print(sum(a))
```

Output: 6

`len(a)` — Returns number of elements/characters.

Example:

```
print(len("hello"))
```

Output: 5

`abs(x)` — Returns the absolute value.

Example:

```
print(abs(-7))
```

Output: 7

`round(x, n)` — Rounds x to n decimal places.

Example:

```
print(round(3.14159, 2))
```

Output: 3.14

`pow(x, y)` — Returns x raised to power y.

Example:

```
print(pow(2,3))
```

Output: 8

`sorted(a)` — Returns a new sorted list.

Example:

```
print(sorted([3,1,2]))
```

Output: [1, 2, 3]

4. TYPE CONVERSION

`int(x)` — Converts to integer.

Example:

```
print(int("123"))
```

Output: 123

`str(x)` — Converts to string.

Example:

```
print(str(123))
```

Output: 123

`float(x)` — Converts to float.

```
Example:
print(float("3.5"))
Output: 3.5
```

`list(x)` — Converts an iterable to a list.

```
Example:
print(list("abc"))
Output: ['a', 'b', 'c']
```

`set(x)` — Creates a set; useful for removing duplicates.

```
Example:
print(set([1,2,2,3]))
Output: {1, 2, 3}
```

5. LOOP HELPERS

`range(n)` — Produces 0 through n-1.

```
Example:
for i in range(3):
    print(i)
Output: 0
1
2
```

`enumerate(a)` — Loops with both index and value.

```
Example:
for i,v in enumerate(["a","b"]):
    print(i,v)
Output: 0 a
1 b
```

`zip(a,b)` — Loops through corresponding elements.

```
Example:
for x,y in zip([1,2],[3,4]):
    print(x,y)
Output: 1 3
2 4
```

`any(iterable)` — True if at least one item is truthy.

```
Example:
print(any([False, True, False]))
Output: True
```

`all(iterable)` — True only if every item is truthy.

```
Example:
print(all([True, True]))
Output: True
```

6. map() AND filter()

`map(function, iterable)` — Applies a function to every item.

```
Example:
print(list(map(int, ["1","2","3"])))
Output: [1, 2, 3]
```

`filter(function, iterable)` — Keeps items for which the function returns True.

```
Example:
print(list(filter(lambda x: x%2==0, [1,2,3,4])))
Output: [2, 4]
```

7. DICTIONARY METHODS

`d.keys()` — Returns dictionary keys.

```
Example:
d={"a":10,"b":20}
print(list(d.keys()))
Output: ['a', 'b']
```

`d.values()` — Returns dictionary values.

```
Example:
d={"a":10,"b":20}
print(list(d.values()))
Output: [10, 20]
```

`d.items()` — Returns key-value pairs.

```
Example:
d={"a":10}
print(list(d.items()))
Output: [('a', 10)]
```

`d.get(key, default)` — Gets a value without KeyError; returns default if missing.

```
Example:
d={"a":10}
print(d.get("b",0))
Output: 0
```